WEEK 3

PROGRAM 1

```
#include <iostream>

using namespace std;

class student

{

private:

   int roll;

   string name;


public:

   void read(int x, string y)

   {

      roll = x;

      name = y;

   }


   void print()

   {

      cout << "Roll no is: " << roll << endl;

      cout << "Name is: " << name;
```

```
    }
};
int main()
{
    student s1;
    int rno;
    string n;
    cout << "Enter roll no and name: ";
    cin >> rno >> n;
    s1.read(rno, n);
    s1.print();
    return 0;
}
```

PROGRAM 2

```
#include <iostream>
using namespace std;

class student {
private:
    int roll;
```

```
    string name;
public:
    student(int x, string y) : roll(x), name(y) {}
    void display() {
        cout << "roll no is: " << roll << endl << "name is: " << name <<
endl;
    }
};


int main() {
    student s1(2, "pragati");
    student s2(4, "kumari");
    s1.display();
    s2.display();
    return 0;
}
```

PROGRAM 3

```
#include <iostream>
using namespace std;
int linearSearch(int arr[], int n, int key) {
    for (int i = 0; i < n; i++) {
```

```
        if (arr[i] == key) {
            return i;
        }
    }
    return -1;
}
int main() {
    int n, key;
    cout << "Enter size of array and key: ";
    cin >> n >> key;

    int arr[n];
    for (int i = 0; i < n; i++) {
        cout << "Enter element " << i+1 << ": ";
        cin >> arr[i];
    }

    int result = linearSearch(arr, n, key);
    if (result != -1)
        cout << "Key found at index " << result;
    else
```

```
        cout << "Element not found";
    return 0;
}
```

PROGRAM 4

```
#include <iostream>
using namespace std;
int main()
{
    int i, n, key, c = 0;
    cout << "Enter size of array and key element ";
    int a[n];
    for (i = 0; i < n; i++)
    {
        cout << "Enter array element " << i+1;
        cin >> a[i];
        if (key == a[i])
        {
            cout << "Key is found at " << i+1;
            c++;
        }
```

```
    }
    cout << "No. of times key repeated = " << c;
    if (c == 0)
        cout << "Key element is not found";

return 0;
}
```

PROGRAM 5

```
#include <iostream>
using namespace std;
int main() {
    int n, key;
    bool status = false;
    cout << "Enter n value and key value: ";
    cin >> n >> key;
    int arr[n];
    int low = 0, high = n - 1;
    cout << "Enter all array elements in sorted order: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
```

```
    }
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key) {
            cout << key << " element found at " << mid << " location";
            status = true;
            break;
        }
        else if (arr[mid] < key) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
    if (!status) {
        cout << "Element not found in the array";
    }
    return 0;
}
```